Working with powers — Simplifying · Index laws · Expanding · Factorising

No calculator. ★ = challenge question.

A · SIMPLIFYING AND INDEX LAWS

1 Simplify

a $9y^2 - 4y^2$ **b** $3x + 2x^3 + 5x - x^3$ **c** $5d^3 - 3 + 2d^3 + 7 - d^2$

> Only like terms (same letter *and* same power) combine. **a** $5y^2$ **b** $8x + x^3$ **c** $7d^3 - d^2 + 4$.

2 Write each expression as a single power.

a $x^5 \times x^4$ **b** $\dfrac{y^{12}}{y^4}$ **c** $(z^3)^4$ **d** $m^2 \times m^6 \times m$ **e** $\dfrac{a^5 \times a^3}{a^6}$

> **a** x^9 **b** y^8 **c** z^{12} **d** $m^{2+6+1} = m^9$ **e** $\dfrac{a^8}{a^6} = a^2$.

3 Simplify

a $4a \times 6b$ **b** $3n \times (-2n)$ **c** $5p^2 \times 2p^3$ **d** $\dfrac{20x^6}{4x^2}$ **e** $(2y)^3$

> Numbers first, then the letters. **a** $24ab$ **b** $-6n^2$ **c** $10p^5$ **d** $5x^4$ **e** $2^3 \times y^3 = 8y^3$.

B · EXPANDING BRACKETS

4 Expand

a $p(p^2 + 3p)$ **b** $3m^2(m - 5)$ **c** $2x(4x^2 - x + 6)$

> Multiply every term inside by the term outside, adding powers. **a** $p^3 + 3p^2$ **b** $3m^3 - 15m^2$ **c** $8x^3 - 2x^2 + 12x$.

5 Expand and simplify

a $4(x + 2) + 3(x - 1)$

b $5(2y - 3) - 2(y + 4)$

c $x(x + 4) + 2x(3x - 1)$

> **a** $4x + 8 + 3x - 3 = 7x + 5$.
>
> **b** $10y - 15 - 2y - 8 = 8y - 23$ (the -2 multiplies both terms in its bracket).
>
> **c** $x^2 + 4x + 6x^2 - 2x = 7x^2 + 2x$.

6 Show that $2x(3x+1)+x(x-2) \equiv 7x^2$.

> Left-hand side: $6x^2+2x+x^2-2x=7x^2$, which is the right-hand side for every value of x — so it is an *identity* ($\equiv$), not just an equation.

C · FACTORISING

7 Factorise

a $3x+15$ **b** $10y-25$ **c** $18-12a$

> Take out the HCF of the terms and check by expanding. **a** $3(x+5)$ **b** $5(2y-5)$ **c** $6(3-2a)$.

8 Factorise completely

a $6y^2+9y$ **b** $8x^3-12x$ **c** $10w^3+15w^2-5w$

> The HCF includes letters too. **a** HCF $3y$: $3y(2y+3)$ **b** HCF $4x$: $4x(2x^2-3)$ **c** HCF $5w$: $5w(2w^2+3w-1)$.

9 Sam factorises $12x^2+18x$ and writes $6(2x^2+3x)$. Explain why this is not factorised *completely*, and write the complete factorisation.

> $2x^2$ and $3x$ still share a factor x, so the bracket can be factorised further. The HCF of $12x^2$ and $18x$ is $6x$: $12x^2+18x=6x(2x+3)$.

D · SUBSTITUTING AND SOLVING

10 Decide whether each statement is an **identity** (true for every value) or an **equation** (true for particular values). If it is an equation, solve it.

a $2x+3x+1=5x+1$

b $4x-2=10$

c $3(y+2)=3y+6$

> **a** Identity — both sides simplify to $5x+1$. **b** Equation: $4x=12$, $x=3$. **c** Identity — expanding the left gives the right.

11 Find the value of each expression when $x=5$, $y=-3$ and $z=2$.

a $3(x-y)+2z$ **b** $2x^2-yz$ **c** $(y+z)^2-x$

> **a** $3(5-(-3))+4=3\times 8+4=28$. **b** $2\times 25-(-3)(2)=50+6=56$. **c** $(-3+2)^2-5=1-5=-4$.

12 Solve $4(x-1)=50-2(x+3)$

> Expand both sides: $4x-4=50-2x-6=44-2x$. Add $2x$: $6x-4=44$. Add 4: $6x=48$, so $x=8$. Check: $4\times 7=28$ and $50-22=28$ ✓

13. A rectangle 4 cm by $(3x - 7)$ cm has the same area as a rectangle 2 cm by $(5x + 2)$ cm. Work out x and the area.

> $4(3x - 7) = 2(5x + 2) \Rightarrow 12x - 28 = 10x + 4 \Rightarrow 2x = 32 \Rightarrow x = 16$. Area $= 4 \times (48 - 7) = 4 \times 41 = 164$ cm² (check: $2 \times 82 = 164$ ✓).

14. The formula $v^2 - u^2 = 2as$ links the starting speed u, the final speed v, the acceleration a and the distance s of a moving object. A car starts from rest ($u = 0$) and accelerates at 1.5 m/s² until it reaches 12 m/s. How far has it travelled?

> Substitute: $12^2 - 0 = 2 \times 1.5 \times s \Rightarrow 144 = 3s \Rightarrow s = 48$ m.

Check: like terms need the same power · index laws need the same base · a bracket times a term — every term inside · "completely" means the HCF, letters included.